

Question	Answer	Marks
4(a)	combined capacitance of parallel capacitors = 30 (μF)	C1
	total capacitance = $(1/45 + 1/30)^{-1}$ = 18 μF	A1
4(b)	$E = \frac{1}{2}CV^2$	C1
	$\Delta E = \frac{1}{2} \times 45 \times 10^{-6} (9.6^2 - 8.0^2)$ = $6.3 \times 10^{-4} \text{ J}$	A1
4(c)(i)	gaps in circuit closed and correct symbol for capacitor shown in parallel with load resistor	B1
4(c)(ii)	two correct pairs of values of t and V read off from within same discharge cycle, e.g. (5.0, 4.0) and (13.0, 3.2)	C1
	correct substitution of values of V , V_0 and Δt into $V = V_0 \exp(-\Delta t / \tau)$ e.g. $3.2 = 4.0 \exp(-8.0 / \tau)$	C1
	$\tau = 36 \text{ ms}$	A1

Question	Answer	Marks
4(d)(i)	8.0 W	A1
4(d)(ii)	4.0 W	A1

Question	Answer	Marks